

Descriptive Analytics Report — Starbucks Sales

One Full Year of Store, Product, Category & Payment-Method Performance Across Four Cities

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Note: This report is built from the supplied Starbucks_Past_Year_Sales_Dataset.csv — 1,825 transaction-level rows covering 365 consecutive days (1 Aug 2025 – 31 Jul 2026), 4 stores (Mumbai, Delhi, Lucknow, Bengaluru), 4 products across 2 categories (Beverage, Bakery), and 3 payment methods (UPI, Card, Cash). Unlike a pre-aggregated snapshot, this dataset has a genuine date field, so — unlike the source template it's based on — this report includes an actual day-by-day and month-by-month trend view rather than a single-period comparison.

1. Objective

- How are total sales trending day-to-day and month-to-month across the year?
- Which products and categories are driving revenue and volume?
- Does demand differ by day of week, or between weekdays and weekends?
- Which stores/cities are the strongest performers, and how does their product mix differ?
- How is revenue split across payment methods, and does that vary by store?

2. How This Would Work at Real Scale

- Starbucks' sales data is generated continuously through customer transactions, so descriptive analytics can be used to monitor sales, products, stores, and payment methods regularly. A realistic architecture:

Layer	Purpose	Typical Tools
Data Ingestion	Capture each transaction including date, store, product, category, quantity, sales and payment method	Python, Excel, CSV
Data Processing	Clean and transform transaction data and calculate daily, monthly, product-wise and store-wise metrics	Python, Pandas
Storage	Store raw transaction records and processed/aggregated sales data for analysis	SQL Database, Cloud Storage
Aggregation	Create summaries such as sales by product, store, category, payment method and month	Python, SQL
Serving / BI Layer	Present dashboards and charts showing sales trends, top products, store performance and payment preferences	Power BI, Tableau, Excel

The tables and charts in this report represent the output of this pipeline — aggregated from transaction-level Starbucks sales data by date, product, category, store and payment method.

3. Market Context: Starbucks (Tata Starbucks) in India

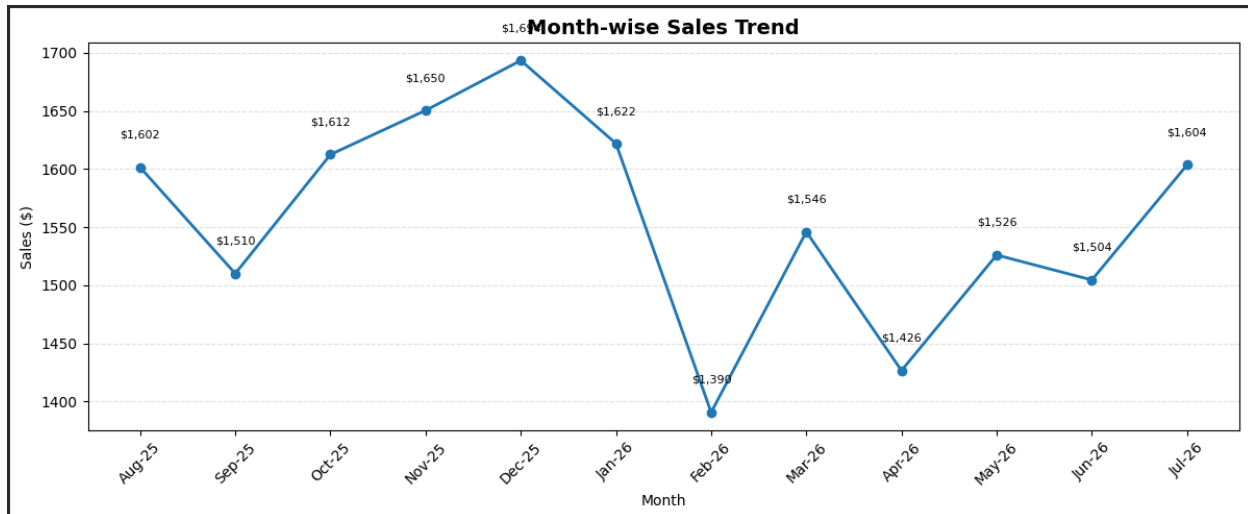
Starbucks operates in India through Tata Starbucks, a 50:50 joint venture between Tata Consumer Products and Starbucks Corporation, founded in 2012. As of FY26 the chain has crossed 500 stores nationally, with 23 net new stores added during the year, and reported ~7% year-on-year revenue growth to ₹1,367 crore (~\$163M) for the 12 months ended March 2026.

Leadership has signalled a more measured, profitability-focused expansion pace going forward — the earlier ambition of 1,000 stores by FY28 has been pushed out — with the business turning EBITDA/EBIT-positive for the first time and same-store sales growth reported as positive for the year. This context is useful for reading the store-level findings below: with expansion now deliberately calibrated, per-store performance and product mix (rather than raw store-count growth) is what actually moves the business.

4. Daily & Monthly Sales Trend

Total sales across the full year: \$18,687 across 4,518 items sold over 1,825 transactions — an average of \$51.20 in sales and exactly 5 transactions per day, with daily sales showing meaningful spread (std. dev. ≈ \$11.68/day) around that average.

Month-by-month sales:



Key Insights

- **December 2025 recorded the highest sales at \$1,693.50**, indicating the strongest monthly performance.
- **February 2026 was the weakest month**, with sales falling to **\$1,390.50** and a **14.3% MoM decline**.
- Sales recovered strongly in **March 2026**, increasing by **11.2%** from February.
- Another noticeable decline occurred in **April 2026 (-7.7%)**, followed by recovery in May.
- **July 2026 sales reached \$1,603.50**, almost matching the starting level of August 2025 (\$1,601.50).
- Overall, sales show **fluctuations rather than a consistent upward or downward trend**.

Overall Interpretation

The monthly sales trend indicates a **relatively stable but fluctuating sales pattern** throughout the year. Sales peaked in December 2025 before experiencing a significant decline in January and February 2026. However, the strong recovery in March and subsequent improvement toward July suggest that the business was able to recover from weaker periods. Overall, the annual sales performance remained stable, with **\$18,687 in total sales and an average monthly sales of \$1,557.25**.

Key Findings

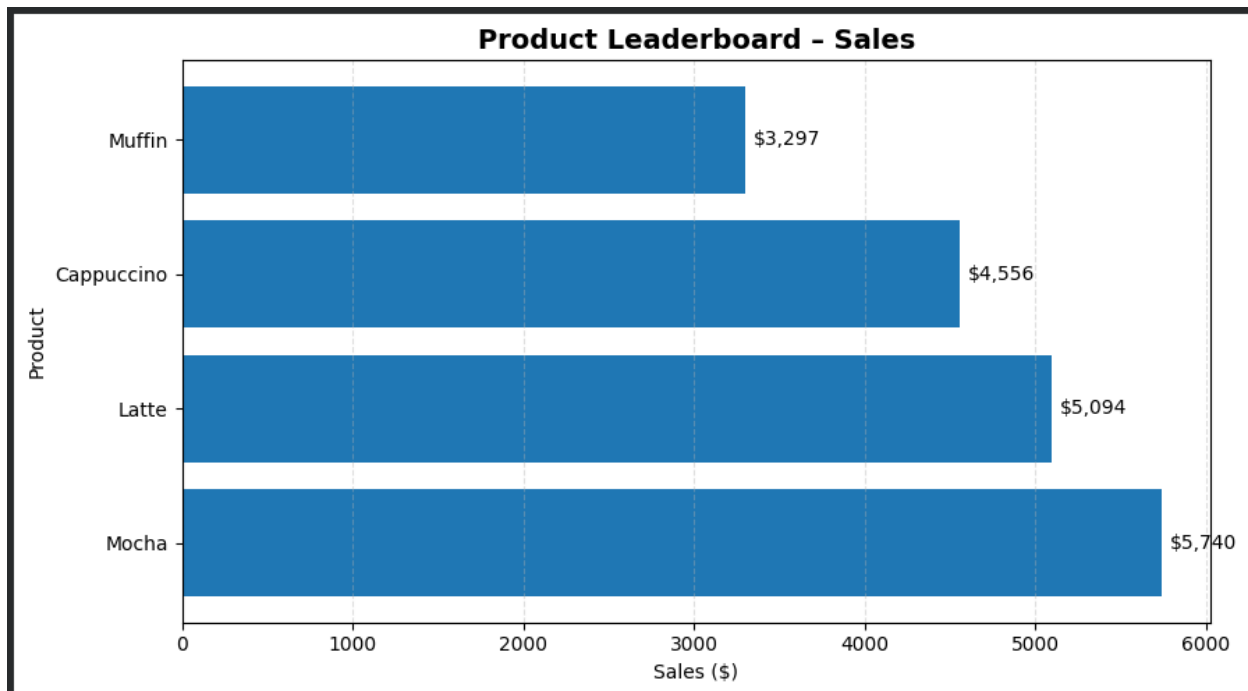
Overall finding:

Sales remained broadly stable throughout the year, but experienced short-term fluctuations, with **December as the strongest month and February as the weakest**. The recovery after February suggests potential seasonal or demand-related fluctuations rather than a long-term decline.

Important business takeaway:

- Investigate why **February sales dropped 14.3%**.
- Identify what drove the **March recovery of 11.2%**.
- Use December's performance to understand successful products/promotions.
- Focus on maintaining the recovery seen toward **June–July 2026**.

5. Product Leaderboard



Key Insights

- **Mocha is the top-performing product**, generating **\$5,740** in sales.
- **Latte ranks second** with **\$5,094**, followed by **Cappuccino at \$4,556**.
- The three beverage products have **very similar transaction counts (450–465)**, but their sales differ because of their different prices.
- **Muffin has 456 transactions**, almost equal to the beverages, but generates only **\$3,297** because its unit price is lower.
- Mocha contributes approximately **30.7% of total sales**, making it the strongest revenue contributor.
- The **three beverage products together contribute about 82.3% of total sales**, showing that beverages dominate the product portfolio.

Overall Interpretation

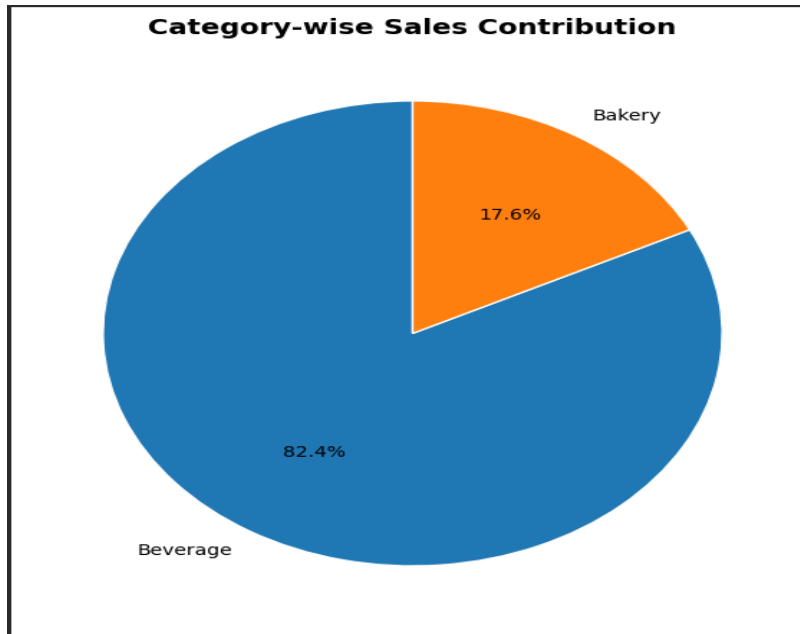
The product leaderboard shows that **beverages are the major revenue drivers**, with Mocha clearly leading in sales, quantity sold, and transactions. Although Muffin has a transaction count comparable to the beverages, its lower price results in significantly lower revenue. This indicates that **product pricing plays an important role in revenue generation**, not just sales volume.

Key Findings

Mocha is the strongest product, generating the highest sales, quantity sold, and transactions. Beverages dominate overall revenue, while Muffin's lower price limits its revenue contribution despite having a similar number of transactions.

Business takeaway: Focus on promoting **Mocha and other high-value beverages**, while exploring ways to increase the **average value of Muffin purchases**, such as combos or cross-selling with beverages.

6. Category Performance — Beverage vs. Bakery



Key Insights

- **Beverages dominate the business**, contributing **82.4% of total sales (\$15,390)**.
- **Bakery contributes only 17.6% (\$3,297)** of total revenue.
- Beverage quantity sold (**3,419**) is more than **3×** the bakery quantity sold (**1,099**).
- Beverage transactions (**1,369**) are also about **3×** bakery transactions (**456**).
- The strong beverage performance is driven by both **higher sales volume and higher-priced products**.

Overall Interpretation

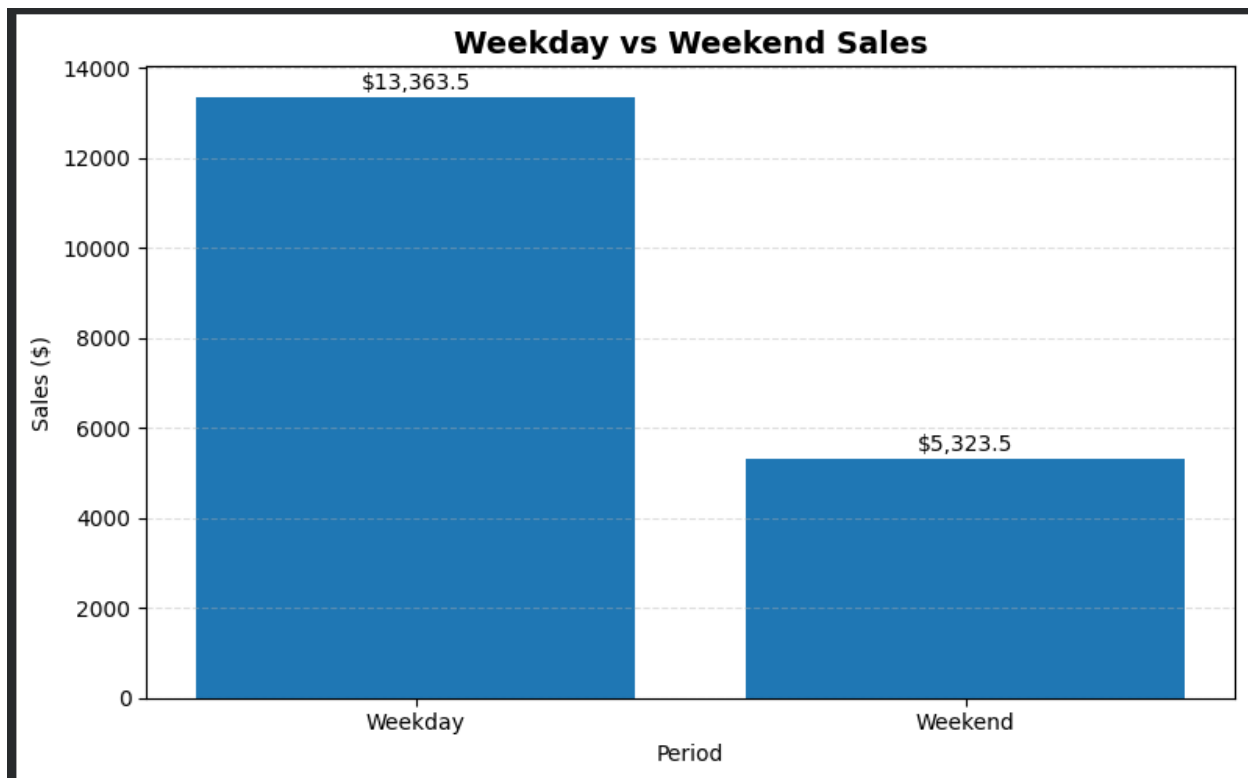
The category analysis shows a **strong dependence on beverages**, which generate more than four-fifths of total sales. Bakery products make a smaller contribution, mainly because the dataset contains only one bakery product (Muffin). Therefore, the category mix indicates that **beverages are the primary revenue driver**, while bakery represents an opportunity for diversification and additional sales.

Key Findings

Beverage is the dominant category, accounting for **82.4% of total sales, 75.7% of quantity sold, and 75.0% of transactions**. Bakery contributes a much smaller share, suggesting that **increasing bakery variety and cross-selling bakery items with beverages could improve overall revenue**.

Business takeaway: Since beverage already performs strongly, the best opportunity is to **maintain beverage sales while increasing bakery sales through combos, add-ons, and product variety**.

7. Weekday vs. Weekend & Day-of-Week Behaviour



Key Insights

- **Weekdays generate 71.5% of total sales**, compared with **28.5% on weekends**.
- Weekdays have much higher transaction volume: **1,305 vs. 520 transactions**.
- Despite the difference in total sales, **average daily sales are almost identical: \$51.20 on weekdays vs. \$51.19 on weekends**.
- This suggests that the lower weekend sales are mainly because there are **fewer weekend days in the dataset**, rather than weaker daily performance.
- There is **no clear weekend sales boost** in this dataset.

Overall Interpretation

The analysis indicates that **weekday and weekend daily sales performance is almost identical**. The higher total weekday sales are primarily explained by the greater number of weekday days and transactions. Therefore, there is **no strong evidence of a weekend-specific spending pattern** in this dataset.

Key Findings

Weekdays contribute more total sales due to higher transaction volume, but average daily sales are virtually identical between weekdays and weekends (\$51.20 vs. \$51.19). This indicates that day type has little impact on daily sales performance in the current dataset.

Business takeaway: Instead of assuming weekends are automatically stronger, the business should analyze **individual days of the week, promotions, store traffic, and transaction-level patterns** to identify genuine demand differences.

KPI: Weekday-to-Weekend Ratio

Value: 2.51x

Usage: Compares total sales performance

Interpretation: Weekday sales are about 2.5x weekend sales

8. Store-Wise (Regional) Performance

Store	Sales (\$)	Qty Sold	Transactions	Avg. Sale (\$)
Bengaluru	4,983.5	1,214	494	10.09
Lucknow	4,732.0	1,135	462	10.24
Mumbai	4,602.0	1,101	435	10.58
Delhi	4,369.5	1,068	434	10.07

Key Insights

- **Bengaluru leads in total sales** with **\$4,983.50** and also has the highest transaction count (**494**).
- **Lucknow ranks second** with **\$4,732**, followed by Mumbai and Delhi.
- **Mumbai has the highest average sale value (\$10.58)** despite having fewer transactions (**435**).
- **Delhi has the lowest sales (\$4,369.50)** and a relatively low transaction count (**434**).
- The difference between the highest and lowest store sales is only **\$614**, showing that regional performance is relatively balanced.
- Mumbai's higher average sale suggests that **product mix or higher-value purchases**, rather than transaction volume, may be driving its performance.

Overall Interpretation

The store-wise analysis shows that **Bengaluru is the strongest store in terms of overall revenue and transaction volume**, while **Mumbai performs best in average transaction value**. Since sales across all four stores are relatively close, there is no major regional underperformer. The differences appear to be driven more by **transaction volume and product mix** than by large differences in overall demand.

Key Findings

Bengaluru is the top-performing store by total sales and transactions, while Mumbai achieves the highest average sale value. Overall, regional sales are relatively balanced, suggesting that improving transaction value and product mix could be more effective than focusing only on increasing customer volume.

Business takeaway: Study **Mumbai's product mix** to identify why its average sale is highest, and apply successful high-value selling strategies to other stores.

9. Store × Product / Category Matrix

Sales (\$) by store and product:

Store	Cappuccino	Latte	Mocha	Muffin
Bengaluru	1,276.0	1,381.5	1,405.0	921.0
Delhi	1,072.0	1,426.5	1,055.0	816.0

Store	Cappuccino	Latte	Mocha	Muffin
Lucknow	1,200.0	1,053.0	1,690.0	789.0
Mumbai	1,008.0	1,233.0	1,590.0	771.0

Key Insights

- Lucknow's Mocha generates the highest individual store-product sales at **\$1,690**.
- Mumbai's Mocha is second highest at **\$1,590**, showing strong demand for Mocha in both cities.
- Delhi performs particularly well with Latte, generating **\$1,426.50**, its highest product sales.
- Muffin is the weakest product across all four stores, with sales ranging from \$771 to \$921.
- Bengaluru has the most balanced product performance, without relying heavily on one product.
- Bengaluru's broad-based performance is consistent with it being the **overall top store**.

Overall Interpretation

The matrix shows that **product preferences differ across stores**. Mocha is particularly strong in Lucknow and Mumbai, while Latte performs best in Delhi. Bengaluru has a more balanced sales distribution across all four products. This suggests that **store-specific product preferences and product mix** play an important role in regional performance.

Key Findings

Lucknow–Mocha is the strongest store-product combination, while Muffin consistently records the lowest sales across stores. Bengaluru's strength comes from balanced performance across the product range rather than dependence on a single product.

Business takeaway: Stores could use **location-specific product strategies**—for example, promoting Mocha more aggressively in Lucknow and Mumbai, while focusing on Latte in Delhi.

10. Payment Method Analysis

Payment Method	Sales (\$)	Transactions	% of Total Sales	Avg. Sale (\$)
Cash	6,453.0	636	34.5%	10.15
UPI	6,149.5	578	32.9%	10.64
Card	6,084.5	611	32.6%	9.96

Key Insights

- Cash leads slightly with **\$6,453**, accounting for **34.5%** of total sales.
- **UPI (32.9%) and Card (32.6%)** are very close to Cash, showing an almost equal three-way split.
- Cash has the highest transaction count at **636 transactions**.
- **UPI has the highest average sale value (\$10.64)**, suggesting UPI users spend slightly more per transaction.
- Card has the **lowest average sale (\$9.96)**, although the difference from UPI is relatively small.
- The difference between the highest and lowest payment-method sales is only **\$368.50**, indicating balanced payment preferences.

Overall Interpretation

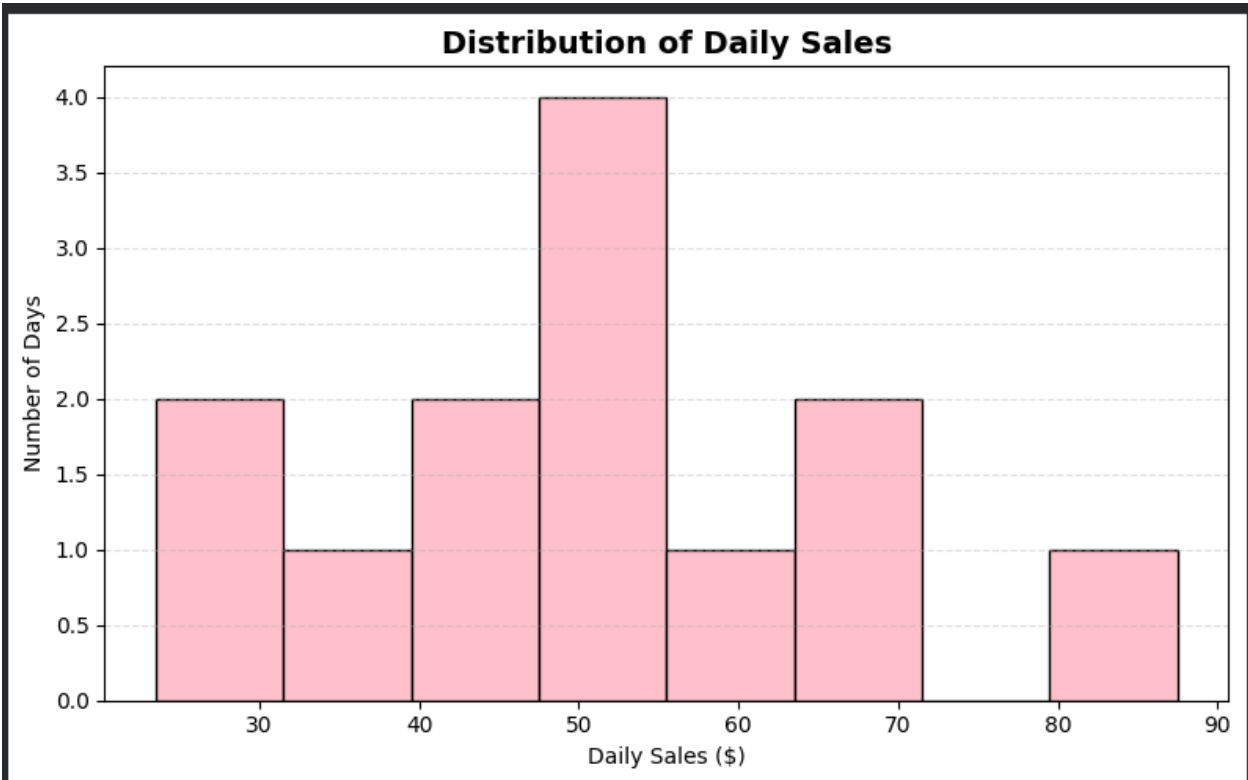
The payment analysis shows that customers use **Cash, UPI, and Card almost equally**, with each contributing roughly one-third of total sales. Cash is marginally dominant in both sales and transaction count, while UPI generates the highest average transaction value. Therefore, **no single payment method dominates customer behavior**.

Key Findings

Payment methods are almost evenly distributed, with Cash contributing the highest sales (34.5%) and UPI recording the highest average sale (\$10.64). The small differences suggest that customers have diverse but relatively balanced payment preferences.

Business takeaway: Continue supporting **all three payment methods**, while monitoring UPI usage because its higher average transaction value could indicate an opportunity for targeted digital-payment offers.

11. Peak & Low Sales Days



Key Insights

- **26 November 2025** was the strongest sales day, generating **\$87.50**.
- **20 September 2025** was the weakest day, with only **\$23.50**.
- Peak-day sales were approximately **3.7× higher** than the lowest day's sales.
- The **\$11.68 standard deviation** indicates noticeable day-to-day sales variation.
- Since only around **5 transactions occur per day**, a few high-value or high-quantity orders can significantly affect daily sales.

Overall Interpretation

The analysis shows **meaningful day-to-day fluctuations in sales**. While the average daily sales are around **\$51.20**, actual daily revenue can vary considerably. This variation may be influenced by transaction size, customer traffic, product mix, promotions, or special events.

Key Findings

Daily sales show considerable variation, with revenue ranging from \$23.50 to \$87.50. The peak day generated nearly 3.7× the sales of the lowest day, indicating that a small number of high-value transactions can substantially influence daily performance.

Business takeaway: Investigate the **26 November peak** and **20 September low** to identify factors such as product mix, promotions, customer traffic, or order size that may explain the difference.

KPI: Average Daily Sales

Value: \$51.20

Usage: Measures the typical daily revenue

Interpretation: A normal day generates around \$51.20

12. Pricing Structure Notes

Unit prices are fixed per product with zero variation across all 1,825 rows: Cappuccino \$4.00, Latte \$4.50, Mocha \$5.00, Muffin \$3.00. There is no discounting, promotional pricing, or price change over the year anywhere in the dataset — every dollar of sales variation comes entirely from quantity and transaction volume, not price movement. This is a meaningful methodological difference from typical retail datasets and should be stated explicitly in any pricing-related conclusions.

13. Data Quality Notes & Limitations

- Fixed transaction count: exactly 5 transactions are recorded every single day for all 365 days with no exceptions — real POS data would show daily transaction counts that vary substantially, so this is very likely a simulated or sampled dataset rather than a full transaction log.
- No true price variation: as noted in Section 12, unit prices never change, so this dataset cannot support any price-elasticity or discount-impact analysis.
- Small product/category range: only 4 SKUs and 2 categories are represented, far narrower than a real Starbucks menu — findings on 'category mix' should be read as illustrative of this sample, not the full India menu.
- No customer identifier: repeat-visit rate, loyalty-program usage, and customer lifetime value cannot be assessed from this data.
- Currency/units: the source file does not label a currency; given the \$3–\$5 unit-price range this report treats values as dollar-denominated for consistency, but this should be confirmed against the assignment brief or original data source before submission.

14. Summary of Key Metrics

Metric	Value
Total Sales (Aug 2025 – Jul 2026)	\$18,687.0
Total Quantity Sold	4,518 items

Metric	Value
Total Transactions	1,825
Average Sale per Transaction	\$10.24
Average Daily Sales	\$51.20
Top Store	Bengaluru — \$4,983.5
Top Product	Mocha — \$5,740.0
Top Category	Beverage — 82.4% of sales
Top Payment Method	Cash — 34.5% of sales
Year-over-Year Growth (Aug→Jul)	+0.1% (effectively flat)
Weekday vs. Weekend Avg. Daily Sales	\$51.20 vs. \$51.19 (no gap)

15. Descriptive → Predictive: Suggested Next Steps

- Sales Forecasting**
 - Predict future daily/monthly sales using historical sales trends.
 - Helps with inventory and revenue planning.
- Demand Prediction**
 - Predict which products will have higher demand in each store.
 - Helps prevent overstocking and stockouts.
- Customer & Transaction Analysis**
 - Predict transaction volume and average order value.
 - Identify factors that influence higher-value purchases.
- Store Performance Prediction**
 - Forecast future sales for Bengaluru, Delhi, Lucknow, and Mumbai.
 - Identify stores with potential for growth or declining performance.
- Peak/Low Day Prediction**
 - Predict high- and low-sales days using factors such as weekday, month, product mix, and promotions.
 - Helps optimize staffing and inventory.
- Promotion Impact Prediction**
 - Use historical sales and discount data to estimate whether a promotion is likely to increase sales.

Appendix: Glossary

- MoM — Month-over-Month change in a metric, expressed as a percentage.
- SKU — Stock Keeping Unit; an individual sellable product (e.g., 'Mocha').
- UPI — Unified Payments Interface, India's real-time interbank payment system.
- SSSG — Same-Store Sales Growth; revenue growth measured only at outlets open for a full comparable period, excluding the effect of new store openings.
- Std. dev. (standard deviation) — a measure of how spread out values are around their average; a higher value means more day-to-day variability.